Applications of Public Key Cryptography

Application of public key crypto systems are classified into three broad categories:

Encryption/Decryption: The sender encrypts the message with the sender's public or private key.

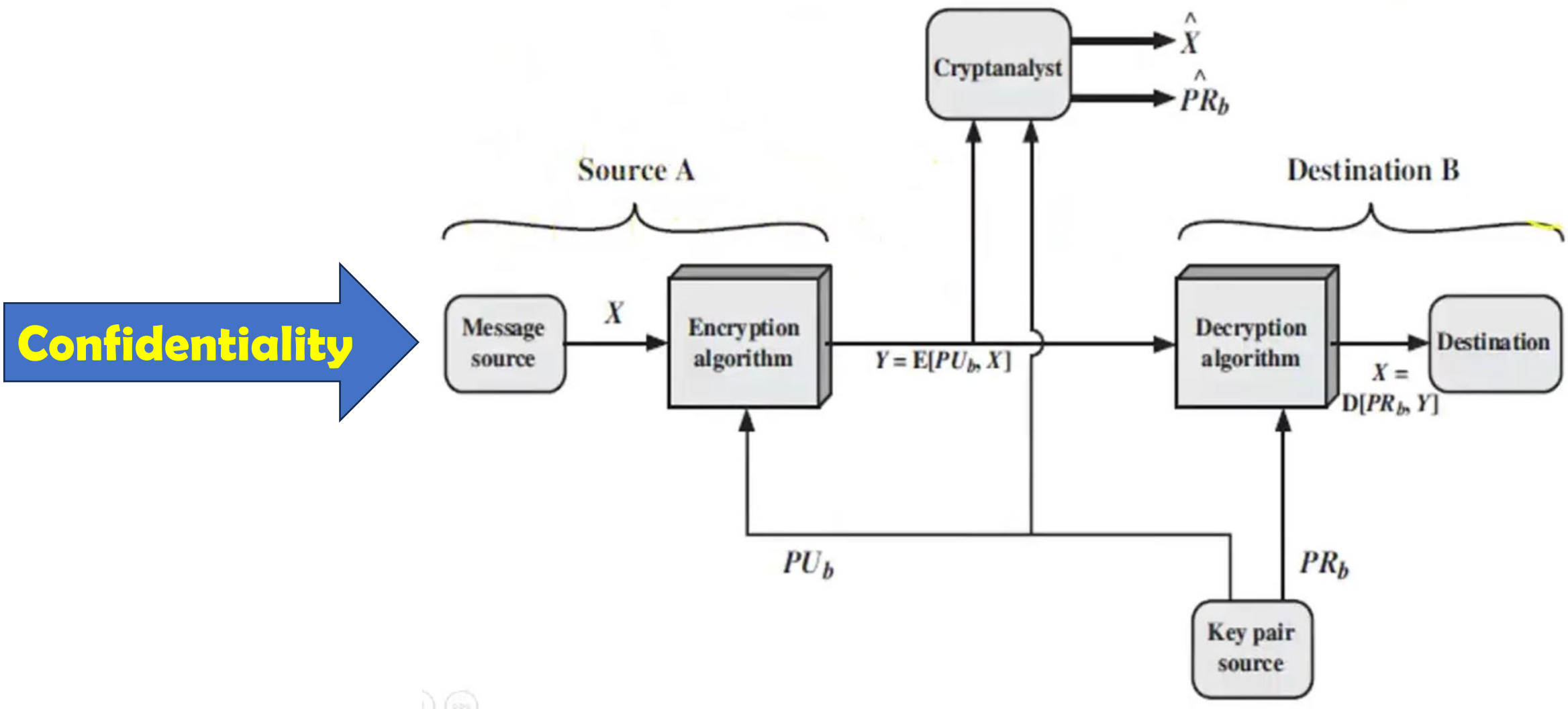
Digital Signature: The sender signs a message with his/her private key.

Key exchange: Both sender and receiver cooperate to exchange their session keys.

- **Confidentiality**
- **Authenticity**

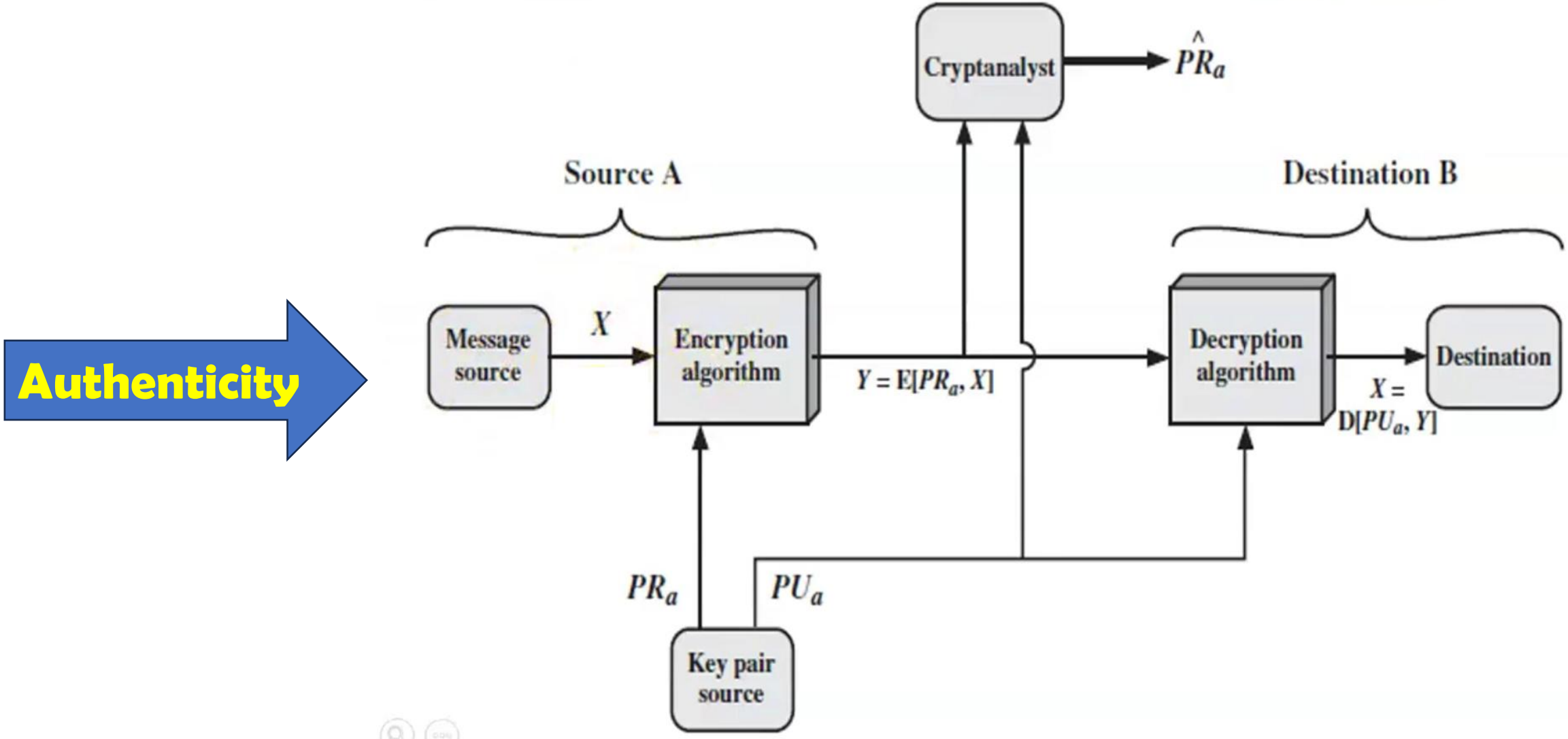
Applications of Public Key Cryptography

- **Confidentiality**



Applications of Public Key Cryptography

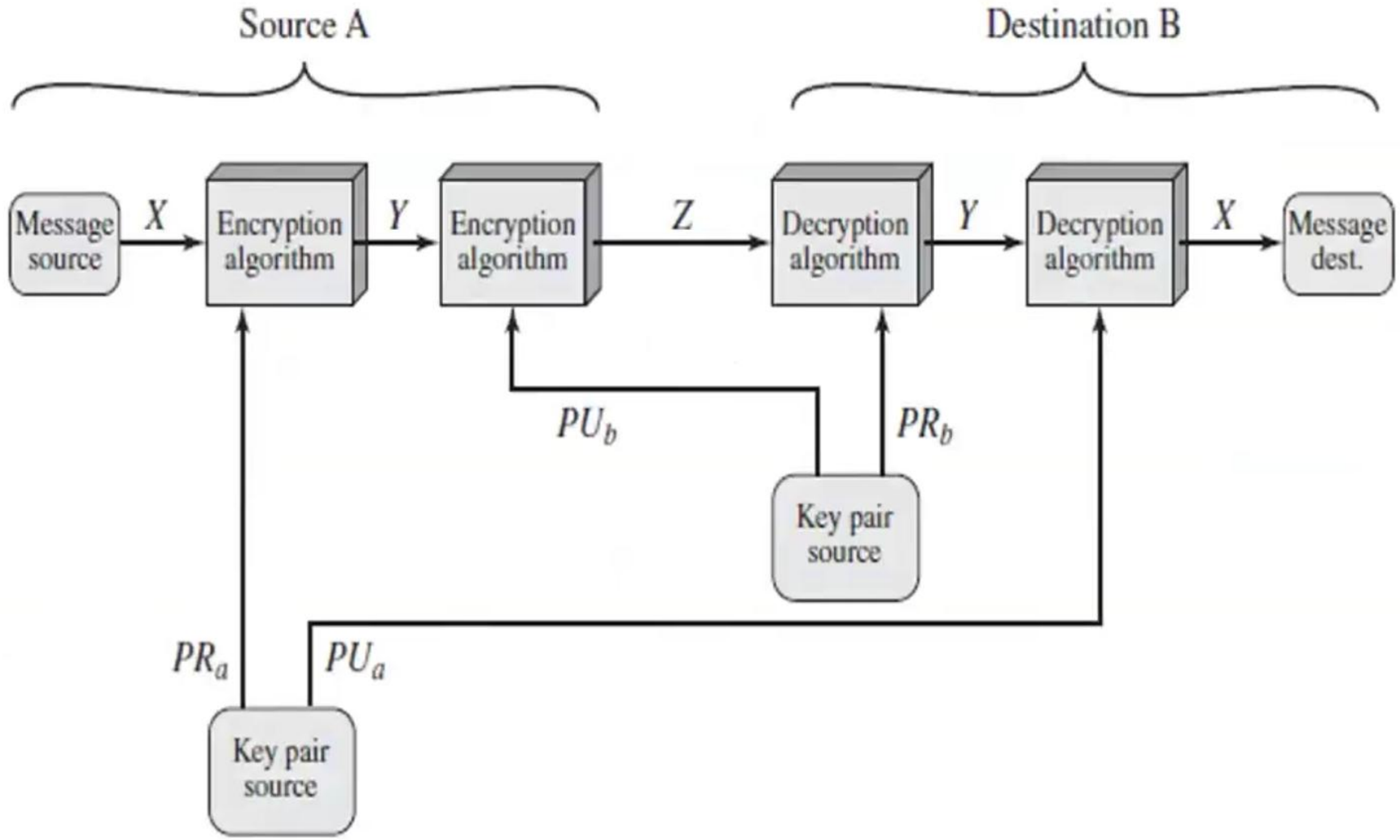
- **Authenticity**



Applications of Public Key Cryptography

- ❑ Confidentiality
- ❑ Authenticity

Authenticity
Confidentiality

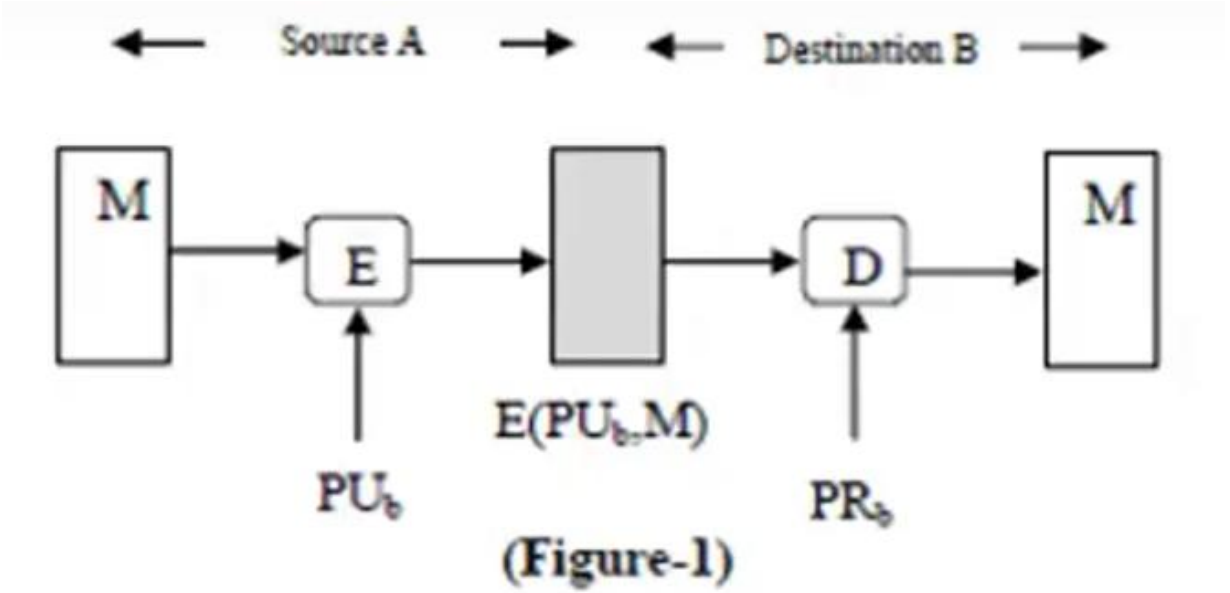


Applications of Public Key Cryptography

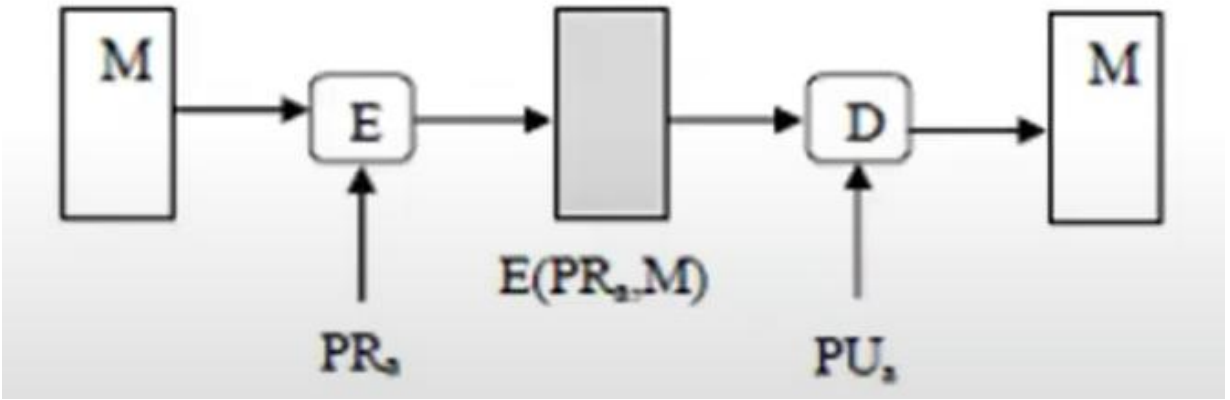
1. Which of the following figures provides authentication only?

2. Which of the following figures provides confidentiality only?

Solutions: Exercise



(Figure-1)



Digital Signature

Digital Signature

A digital signature is a mathematical technique used to validate the authenticity and integrity of a message or digital document.

A digital signature generated electronically from the digital computer to ensure the identity of the sender and the content of the message cannot be modified during the transmission process.



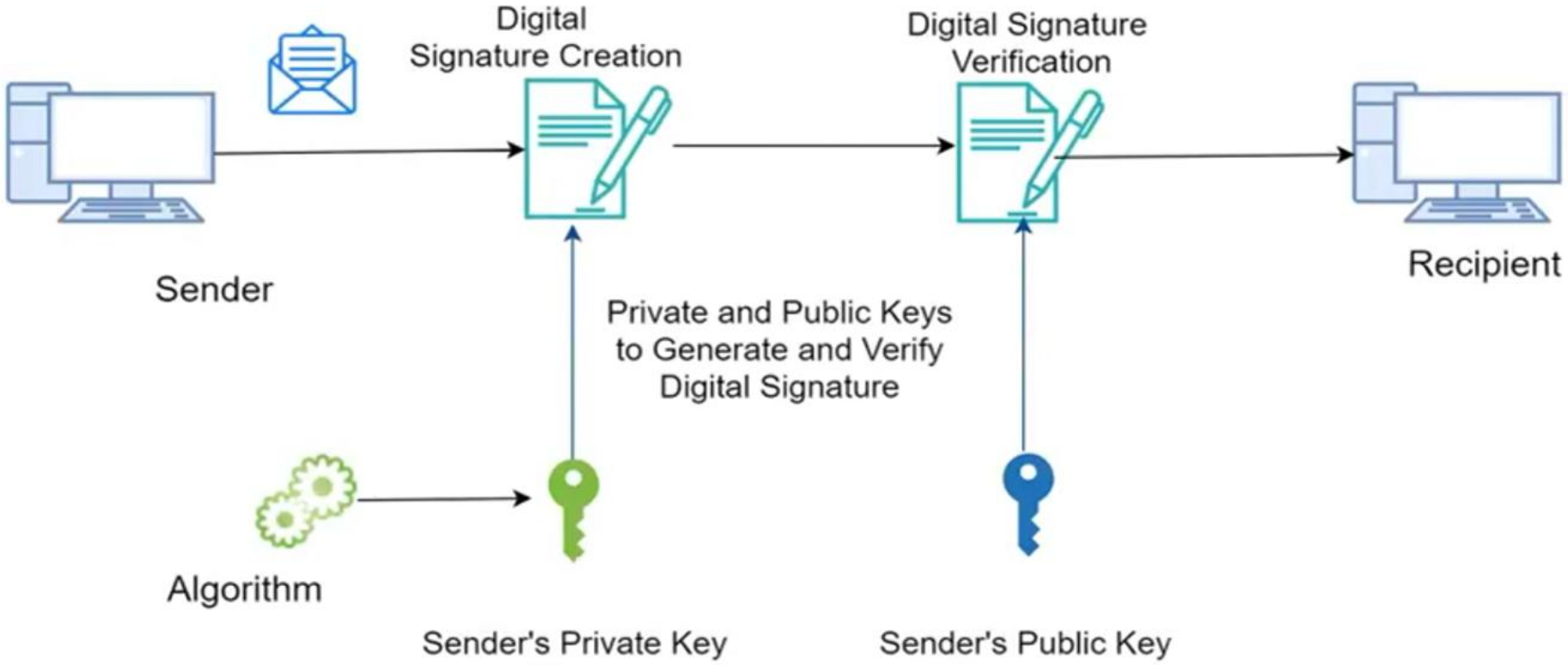


Purpose of Digital Signature

Concept of digital signature is that sender of a message uses a signing key to sign the message and send that message. Its digital signature.

The receiver uses a verification key (public key) of the sender only to verify the origin of the message and make sure that it has not been tempered with during the transmission.

Digital signature technique achieves the authenticity and integrity of the data over internet.



Process of Digital Signature

Hash value of a message when encrypted with the private key of the sender is his/her digital signature on e-document. Three different algorithms are used to complete the process of digital signature.

1. A key generation algorithm is used to generate the private key and corresponding public key.
2. A signing algorithm selects the message and the private key of the sender to produce a signature.



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1. A key generation algorithm is used to generate the private key and corresponding public key.
2. A signing algorithm selects the message and the private key of the sender to produce a signature.
3. A verification algorithm verifies the authenticity and integrity of the sender and the message .



Digital Signature

Uses and Applications of digital signature???